

PHY 412: Statistical Physics

25-26 Semester II

Assignment – 5

Problems in black are for submission; Blue ones are for practice. Due date: April 24th in Class

1. Consider one particle in quantum canonical ensemble in a cubic volume V with periodic boundary conditions.
 - (a) Find the partition function Z .
 - (b) Find $\langle x | \rho | x' \rangle$ explicitly using $\rho = \frac{\sum_n \exp(-\beta E_n) |E_n\rangle \langle E_n|}{Z}$ and the coordinate representation of $|E_n\rangle$.
 - (c) Find ρ in the momentum representation.
2. Consider an electron in a magnetic field $\mathbf{B}$ in the quantum canonical ensemble. Its Hamiltonian is $-\mu_B \hat{\sigma} \cdot \mathbf{B}$.
 - (a) Find the density matrix if $\mathbf{B}$ points along y-axis, and
 - (b) Find the average energy of the system using the density matrix.
3. We know that any many-body quantum system of identical particles is described by a Hamiltonian that is invariant under the interchange of all the coordinates (position, spin, etc.) of any two particles. Prove that any wave function of the system cannot be written as the linear superposition of the combination of Hamiltonian eigen Kets such that some eigen Kets are symmetric and some are antisymmetric.
4. Using grand canonical ensemble, show that for the Bose system and the Fermi system
 - (a) $\langle n_\epsilon^2 \rangle - \langle n_\epsilon \rangle^2 = \langle n_\epsilon \rangle \pm \langle n_\epsilon \rangle^2$ (Which sign is for which system?)
 - (b) $\langle n_{\epsilon_i} n_{\epsilon_j} \rangle = \langle n_{\epsilon_i} \rangle \langle n_{\epsilon_j} \rangle$ for $i \neq j$.
 - (c) $\langle n_{\text{cell}}^2 \rangle - \langle n_{\text{cell}} \rangle^2 = \langle n_{\text{cell}} \rangle \pm \sum_{\epsilon \in \text{cell}} \langle n_\epsilon \rangle^2$. Here $n_{\text{cell}} \equiv \sum_{\epsilon \in \text{cell}} n_\epsilon$ is the number of particles in a 'cell' consisting of a bunch of nearby energy states.
5. Consider a gas of three identical, non-interacting, free particles of mass m (ignore spin) in a large volume V . Assuming the particles to be either Fermions or Bosons, find the leading order (quantum) correction to the specific heat C_V (determined using Maxwell-Boltzmann statistics) in the limit of very high temperature.
6. Find the chemical potentials of the Fermi and the Bose ideal 2D gases exactly in terms of temperature and particle (areal) density. Argue Bose-Einstein condensation is not possible in 2D gases at any finite temperature.
7. Consider a gas of N noninteracting electrons in a uniform external magnetic field, B , at $T \gg T_F$. Let the magnetic moment of an electron be $\mu_B = |e|\hbar/2m$. Show that magnetic susceptibility, $\chi = N\mu_B^2/k_B T$ (Pauli paramagnetism) by explicitly finding magnetization, $M = \mu_B(N_\uparrow - N_\downarrow)$, where $N_\uparrow$ and $N_\downarrow$ are the numbers of electrons with up spin and with down spin respectively.